

YINGBING HUANG

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EDUCATION

- **University of Illinois at Urbana-Champaign**, Urbana, IL **Aug. 2023 - Present**
Ph.D. Student in Electrical and Computer Engineering, **GPA: 3.89/4.0**
Advisor: Deming Chen
- **Cornell University**, Ithaca, NY **Aug. 2019 - May. 2023**
B.S. with Honors in Computer Science (**Cum Laude**, 3.76/4.0)
- **Tsinghua University**, Beijing, China **Aug. 2020 - May. 2021**
Exchange Student in Computer Science
- **Columbia University**, New York, NY **July. 2018 - Aug. 2018**
Summer School Student in Data Science and Machine Learning.

WORK EXPERIENCES

- **University of Illinois at Urbana-Champaign**, Urbana, IL *Aug. 2023 - Present*
Research Assistant, Electrical and Computer Engineering Department
Advisor: Prof. Deming Chen
- **University of Washington**, Seattle, WA *Sep. 2022 - May 2023*
Research Assistant, Computer Science Department
Advisor: Prof. Guanya Shi
- **MunichRe Life Insurance**, New York, NY *May. 2022 - Aug. 2022*
Full-time Intern, Data and Machine Learning Scientist
Advisor: Magd Bayoumi
- **Microsoft**, Beijing, China *Jan. 2021 - July. 2021*
Full-time Intern, Machine Learning Inference
Advisor: Wen Yang
- **Cornell University**, Ithaca, NY *Sep. 2021 - May. 2023*
Research Assistant, Reinforcement Learning
Advisor: Prof. Wen Sun
- **TGood**, Qingdao, China *Dec. 2020 - Jan. 2021*
Full-time Intern, Machine Learning
Advisor: Luda Xu

PUBLICATIONS

- [1] **JAM: Controllable and Responsible Text Generation via Causal Reasoning and Latent Vector Manipulation**
Yingbing Huang, Deming Chen, Abhishek K. Umrawal
Link: [arxiv](#)
- [2] **[NeurIPS'24] SnapKV: LLM Knows What You are Looking for Before Generation**
Yingbing Huang*, Yuhong Li*, Bowen Yang, Bharat Venkitesh, Acyr Locatelli, Hanchen Ye, Tianle Cai, Patrick Lewis, and Deming Chen (*equal contribution)
Link: [snapkv.github](#)
The Thirty-ninth Annual Conference on Neural Information Processing Systems (NeurIPSs), 2024

- [3] [DAC'24] **New Solutions on LLM Acceleration, Optimization, and Application (Invited)**
Yingbing Huang, Lily Jiabin Wan, Hanchen Ye, Manvi Jha, Jinghua Wang, Yuhong Li, Xiaofan Zhang,
 and Deming Chen
The ACM/IEEE Design Automation Conference (DAC), 2024
- [4] [ASP-DAC'24] **Software/Hardware Co-design for LLM and Its Application for Design Verification (Invited)**
Yingbing Huang*, Lily Jiabin Wan*, Yuhong Li, Hanchen Ye, Jinghua Wang, Xiaofan Zhang, and Deming Chen (*equal contribution)
The Asia and South Pacific Design Automation Conference (ASP-DAC), 2024
- [5] [ICLR'24] **Adversarial Imitation Learning via Boosting**
 Jonathan D Chang, **Yingbing Huang***, Dhruv Sreenivas*, Kianté Brantley, Wen Sun (*equal contribution)
The Twelfth International Conference on Learning Representations (ICLR), 2024
- [6] [NeurIPS'24] **Active representation learning for general task space with applications in robotics**
 Yifang Chen, **Yingbing Huang**, Simon S Du, Kevin G Jamieson, Guanya Shi
The Thirty-eighth Annual Conference on Neural Information Processing Systems (NeurIPS), 2023

SELECTED PROJECTS

- **Efficient LLM Reasoning** *April. 2025 - Present*
 - Design a fine-tuning strategy to improve the reasoning efficiency of LLMs by utilizing latent representation.
- **Controllable Generation in LLMs** *May. 2024 - Present*
 - Deliver a cause-effect analysis on LLMs with counterfactual evaluation.
 - Design a method to control LLMs generation to align human preference.
- **Cornell Bias Lab, Team Lead** *Jan. 2021 - May. 2022*
 - Investigated the bias existing in multi-modal LLMs such as DALL-E and CLIP.
 - Designed a method to mitigate the bias, including genders, races, occupations, etc.
 - Organized two public events with more than 50 people participated to present the results and draw public attention to bias and fairness in machine learning models.
- **Data Analysis on Obituaries on the New York Times** *Sep. 2020 - May. 2021*
 - Investigated what kind of people could be on the obituaries on New York Times based on location, gender, race, achievement, etc.
 - Built an algorithm to better extract the location in the text, and analyze the relationship and change overtime among different dimensions.

AWARDS AND SCHOLARSHIPS

- **UIUC James M. Henderson Fellowship** *Sep. 2024*
- **Cornell Cum Laude (Top 15%)** *May. 2023*
- **Cornell Dean's List** *Dec. 2019 - May. 2023*

TEACHING SERVICES

- **Introduction to Computing, Teaching Assistant**
 UIUC ECE 120 *Fall 2024*
- **Algorithm, Teaching Assistant**
 UIUC ECE 374 *Summer 2024*
- **Introduction to Reinforcement Learning, Teaching Assistant**
 Cornell CS 4789 *Spring 2023*

- **Introduction to Machine Learning, Teaching Assistant**

Cornell CS 4780

Spring, Fall 2022

- **Linear Algebra, Teaching Assistant**

Cornell MATH 4310

Fall 2021

SELECTED COURSES

- **University of Illinois Urbana Champaign**

ECE 543 Statistical Learning

CS 546 Advanced Topics in NLP

ECE 527 System-On-Chip Design

- **Cornell University**

CS 5412 Cloud Computing

CS 5789 Advanced Topics in Reinforcement Learning

CS 4670 Computer Vision

CS 4410 Operating Systems

CS 4220 Numerical Analysis

ORIE 4580 Stimulation Modeling & Analysis

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, R, Julia, Ocaml, C++, Verilog, etc.
- **Tools:** Pytorch, Pandas, Wandb, Tableau, Jupyter Notebook, TensorFlow, Azure, Vivado/Vitis HLS, $\LaTeX$, etc.